

Numerical Examples: Maxwells equations and Time Varying Fields

Example-1

A conducting bar can slide freely over two conducting rails as shown in Figure 1. Calculate the induced voltage in the bar

- (a) If the bar is stationed at $y = 8$ cm and

$$\mathbf{B} = 4 \cos 10^6 t \mathbf{a}_z \text{ mWb/m}^2$$

- (b) If the bar slides at a velocity

$$\mathbf{u} = 20 \mathbf{a}_y \text{ m/s}$$

and

$$\mathbf{B} = 4 \mathbf{a}_z \text{ mWb/m}^2$$

- (c) If the bar slides at a velocity

$$\mathbf{u} = 20 \mathbf{a}_y \text{ m/s}$$

and

$$\mathbf{B} = 4 \cos(10^6 t - y) \mathbf{a}_z \text{ mWb/m}^2$$

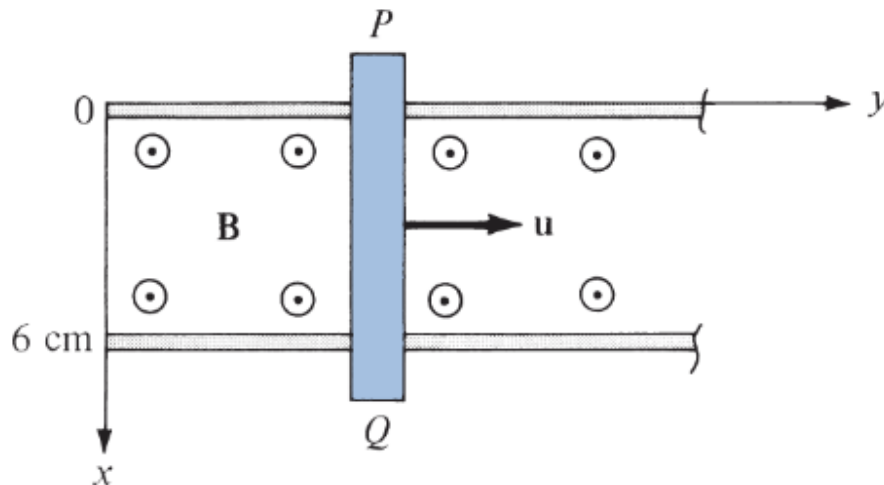


Figure 1: A conducting bar

Solution:

(a) In this case, we have transformer emf given by

$$\begin{aligned}
 V_{\text{emf}} &= - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = \int_{y=0}^{0.08} \int_{x=0}^{0.06} 4(10^{-3})(10^6) \sin 10^6 t \, dx \, dy \\
 &= 4(10^3)(0.08)(0.06) \sin 10^6 t \\
 &= 19.2 \sin 10^6 t \, \text{V}
 \end{aligned}$$

The polarity of the induced voltage (according to Lenz's law) is such that point P on the bar is at lower potential than Q when $\mathbf{B}$ is increasing.

(b) This is the case of motional emf:

$$\begin{aligned}
 V_{\text{emf}} &= \int_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = \int_{x=\ell}^0 (u\mathbf{a}_y \times B\mathbf{a}_z) \cdot dx \mathbf{a}_x \\
 &= -uB\ell = -20(4 \times 10^{-3})(0.06) \\
 &= -4.8 \, \text{mV}
 \end{aligned}$$

(c) Both transformer emf and motional emf are present in this case.

$$V_{\text{emf}} = -\frac{\partial \Psi}{\partial t}$$

where

$$\begin{aligned}
 \Psi &= \int \mathbf{B} \cdot d\mathbf{S} \\
 &= \int_{y=0}^y \int_{x=0}^{0.06} 4 \cos(10^6 t - y) \, dx \, dy \\
 &= -4(0.06) \sin(10^6 t - y) \Big|_{y=0}^y \\
 &= -0.24 \sin(10^6 t - y) + 0.24 \sin 10^6 t \, \text{mWb}
 \end{aligned}$$

But

$$\frac{dy}{dt} = u \rightarrow y = ut = 20t$$

Hence,

$$\Psi = -0.24 \sin(10^6 t - 20t) + 0.24 \sin 10^6 t \, \text{mWb}$$

$$\begin{aligned}
V_{\text{emf}} &= -\frac{\partial \Psi}{\partial t} = 0.24(10^6 - 20) \cos(10^6 t - 20t) - 0.24(10^6) \cos 10^6 t \text{ mV} \\
&\simeq 240 \cos(10^6 t - y) - 240 \cos 10^6 t \text{ V}
\end{aligned} \tag{9.1.5}$$

The dependence of y on time is taken care of in

$$\int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l},$$

and we should not be bothered by it in $\partial \mathbf{B} / \partial t$. Why? Because in computing the transformer emf, the loop is assumed stationary.

Example2:

The loop shown in Figure 2 is inside a uniform magnetic field

$$\mathbf{B} = 50 \mathbf{a}_x \text{ mWb/m}^2.$$

If side DC of the loop cuts the flux lines at the frequency of 50 Hz and the loop lies in the yz -plane at time $t = 0$, find

- (a) The induced emf at $t = 1 \text{ ms}$
- (b) The induced current at $t = 3 \text{ ms}$

Solution:

Solution:

- (a) Since the $\mathbf{B}$ field is time invariant, the induced emf is motional, that is,

$$V_{\text{emf}} = \int_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

where

$$d\mathbf{l} = d\mathbf{l}_{DC} = dz \mathbf{a}_z, \quad \mathbf{u} = \frac{d\mathbf{l}'}{dt} = \rho \frac{d\phi}{dt} \mathbf{a}_\phi = \rho \omega \mathbf{a}_\phi$$

$$\rho = AD = 4 \text{ cm}, \quad \omega = 2\pi f = 100\pi$$

Because $\mathbf{u}$ and $d\mathbf{l}$ are in cylindrical coordinates, we transform $\mathbf{B}$ into cylindrical coordinates:

$$\mathbf{B} = B_0 \mathbf{a}_x = B_0 (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi)$$

where $B_0 = 0.05$. Hence,

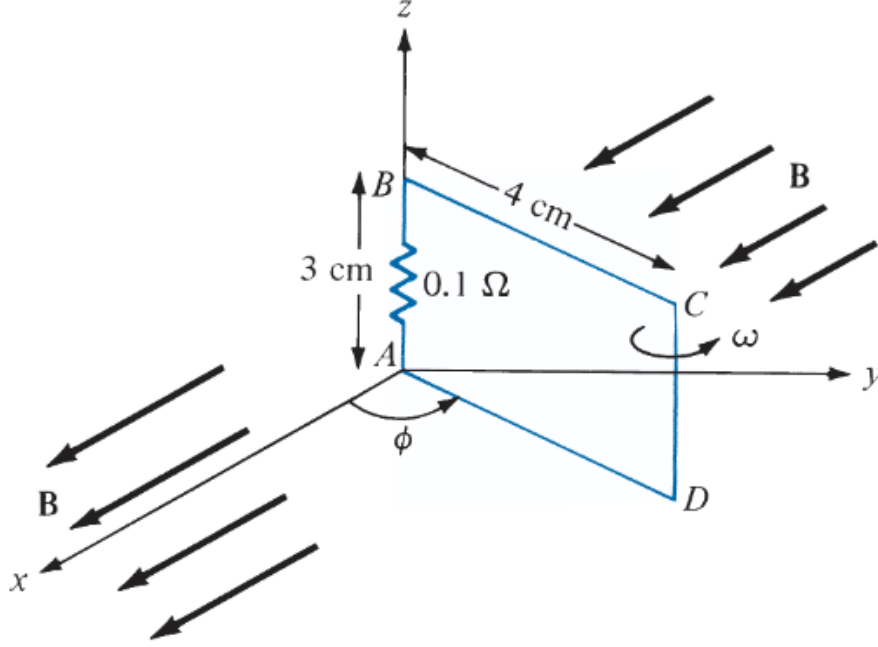


Figure 2: polarity is for increasing emf.

$$\mathbf{u} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ 0 & \rho\omega & 0 \\ B_0 \cos \phi & -B_0 \sin \phi & 0 \end{vmatrix} = -\rho\omega B_0 \cos \phi \mathbf{a}_z$$

and

$$\begin{aligned} (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} &= -\rho\omega B_0 \cos \phi dz = -0.04(100\pi)(0.05) \cos \phi dz \\ &= -0.2\pi \cos \phi dz \end{aligned}$$

$$V_{\text{emf}} = \int_{z=0}^{0.03} -0.2\pi \cos \phi dz = -6\pi \cos \phi \text{ mV}$$

To determine ϕ , recall that

$$\omega = \frac{d\phi}{dt} \rightarrow \phi = \omega t + C_0$$

where C_0 is an integration constant. At $t = 0$, $\phi = \pi/2$ because the loop is in the yz -plane at that time, $C_0 = \pi/2$. Hence,

$$\phi = \omega t + \frac{\pi}{2}$$

and

$$V_{\text{emf}} = -6\pi \cos\left(\omega t + \frac{\pi}{2}\right) = 6\pi \sin(100\pi t) \text{ mV}$$

At $t = 1 \text{ ms}$,

$$V_{\text{emf}} = 6\pi \sin(0.1\pi) = 5.825 \text{ mV}$$

(b) The current induced is

$$i = \frac{V_{\text{emf}}}{R} = 60\pi \sin(100\pi t) \text{ mA}$$

At $t = 3 \text{ ms}$,

$$i = 60\pi \sin(0.3\pi) \text{ mA} = 0.1525 \text{ A}$$

Example 3:

The magnetic circuit of Figure 3 has a uniform cross section of 10^{-3} m^2 . If the circuit is energized by a current

$$i_1(t) = 3 \sin 100\pi t \text{ A}$$

in the coil of $N_1 = 200$ turns, find the emf induced in the coil of $N_2 = 100$ turns. Assume that

$$\mu = 500 \mu_0.$$

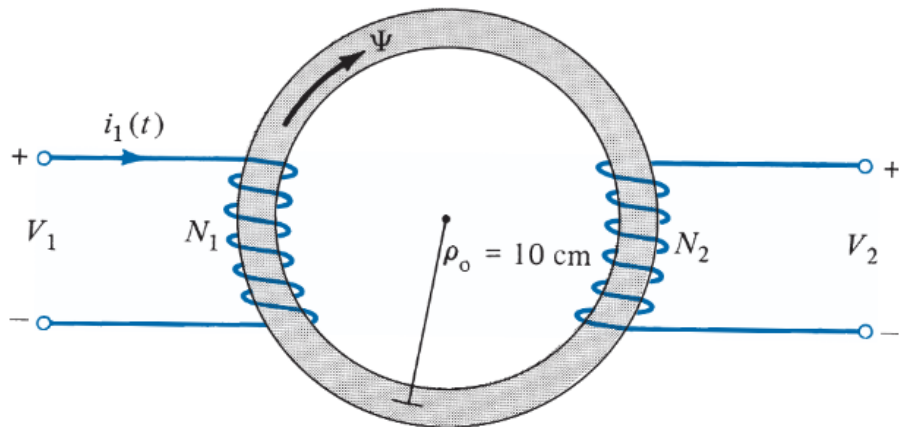


Figure 3: Magnetic circuit

Solution:

The flux in the circuit is

$$\Psi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{N_1 i_1}{\ell/\mu S} = \frac{N_1 i_1 \mu S}{2\pi \rho_0}$$

According to Faraday's law, the emf induced in the second coil is

$$\begin{aligned} V_2 &= -N_2 \frac{d\Psi}{dt} = -\frac{N_1 N_2 \mu S}{2\pi \rho_0} \frac{di_1}{dt} \\ &= -\frac{100 \cdot (200) \cdot (500) \cdot (4\pi \times 10^{-7}) \cdot (10^{-3}) \cdot 300\pi \cos 100\pi t}{2\pi(10 \times 10^{-2})} \\ &= -6\pi \cos 100\pi t \text{ V} \end{aligned}$$

Example 4:

A parallel-plate capacitor with plate area of 5 cm² and plate separation of 3 mm has a voltage

$$50 \sin 10^3 t \text{ V}$$

applied to its plates. Calculate the displacement current assuming

$$\varepsilon = 2\varepsilon_0.$$

Solution:

$$\mathbf{D} = \varepsilon \mathbf{E} = \varepsilon \frac{V}{d}$$

$$J_d = \frac{\partial \mathbf{D}}{\partial t} = \frac{\varepsilon}{d} \frac{dV}{dt}$$

Hence,

$$I_d = J_d \cdot S = \frac{\varepsilon S}{d} \frac{dV}{dt} = C \frac{dV}{dt}$$

which is the same as the conduction current, given by

$$I_c = \frac{dQ}{dt} = S \frac{d\rho_s}{dt} = S \frac{dD}{dt} = \varepsilon S \frac{dE}{dt} = \frac{\varepsilon S}{d} \frac{dV}{dt} = C \frac{dV}{dt}$$

$$I_d = 2 \cdot \frac{10^{-9}}{36\pi} \cdot \frac{5 \times 10^{-4}}{3 \times 10^{-3}} \cdot 10^3 \times 50 \cos 10^3 t$$

$$= 147.4 \cos 10^3 t \text{ nA}$$

Example 5:

Given that

$$\mathbf{A} = 10 \cos(10^8 t - 10x + 60^\circ) \mathbf{a}_z$$

and

$$\mathbf{B}_s = \left(\frac{20}{j}\right) \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y,$$

express $\mathbf{A}$ in phasor form and $\mathbf{B}_s$ in instantaneous form.

Solution:

$$\mathbf{A} = \text{Re} [10e^{j(\omega t - 10x + 60^\circ)} \mathbf{a}_z]$$

where $\omega = 10^8$. Hence

$$\mathbf{A} = \text{Re} [10e^{j(60^\circ - 10x)} \mathbf{a}_z e^{j\omega t}] = \text{Re} (\mathbf{A}_s e^{j\omega t})$$

or

$$\mathbf{A}_s = 10e^{j(60^\circ - 10x)} \mathbf{a}_z$$

If

$$\begin{aligned} \mathbf{B}_s &= \frac{20}{j} \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y = -j20 \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y \\ &= 20e^{-j\pi/2} \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y \end{aligned}$$

$$\mathbf{B} = \text{Re} (\mathbf{B}_s e^{j\omega t})$$

$$\begin{aligned} &= \text{Re} [20e^{j(\omega t - \pi/2)} \mathbf{a}_x + 10e^{j(\omega t + 2\pi x/3)} \mathbf{a}_y] \\ &= 20 \cos \left(\omega t - \frac{\pi}{2} \right) \mathbf{a}_x + 10 \cos \left(\omega t + \frac{2\pi x}{3} \right) \mathbf{a}_y \\ &= 20 \sin \omega t \mathbf{a}_x + 10 \cos \left(\omega t + \frac{2\pi x}{3} \right) \mathbf{a}_y \end{aligned}$$

Example 6

The electric field and the magnetic field in free space are given by

$$\mathbf{E} = \frac{50}{\rho} \cos(10^6 t + \beta z) \mathbf{a}_\phi \text{ V/m}$$

$$\mathbf{H} = \frac{H_0}{\rho} \cos(10^6 t + \beta z) \mathbf{a}_\rho \text{ A/m}$$

Express these in phasor form and determine the constants H_0 and β such that the fields satisfy Maxwell's equations.

Solution:

The instantaneous forms of $\mathbf{E}$ and $\mathbf{H}$ are written as

$$\mathbf{E} = \text{Re}(\mathbf{E}_s e^{j\omega t}), \quad \mathbf{H} = \text{Re}(\mathbf{H}_s e^{j\omega t})$$

where $\omega = 10^6$ and phasors $\mathbf{E}_s$ and $\mathbf{H}_s$ are given by

$$\mathbf{E}_s = \frac{50}{\rho} e^{j\beta z} \mathbf{a}_\phi, \quad \mathbf{H}_s = \frac{H_0}{\rho} e^{j\beta z} \mathbf{a}_\rho$$

For free space,

$$\rho_v = 0, \quad \sigma = 0, \quad \varepsilon = \varepsilon_0, \quad \text{and} \quad \mu = \mu_0,$$

so Maxwell's equations become

$$\nabla \cdot \mathbf{D} = \varepsilon_0 \nabla \cdot \mathbf{E} = 0 \rightarrow \nabla \cdot \mathbf{E}_s = 0$$

$$\nabla \cdot \mathbf{B} = \mu_0 \nabla \cdot \mathbf{H} = 0 \rightarrow \nabla \cdot \mathbf{H}_s = 0$$

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \rightarrow \nabla \times \mathbf{H}_s = j\omega \varepsilon_0 \mathbf{E}_s$$

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \rightarrow \nabla \times \mathbf{E}_s = -j\omega \mu_0 \mathbf{H}_s$$

From the following two equations two Maxwell's equations are satisfied; that is,

$$\nabla \cdot \mathbf{E}_s = \frac{1}{\rho} \frac{\partial}{\partial \phi} (E_{\phi s}) = 0$$

$$\nabla \cdot \mathbf{H}_s = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho H_{\rho s}) = 0$$

Now

$$\nabla \times \mathbf{H}_s = \nabla \times \left(\frac{H_0}{\rho} e^{j\beta z} \mathbf{a}_\rho \right) = \frac{jH_0\beta}{\rho} e^{j\beta z} \mathbf{a}_\phi \quad (9.7.7)$$

We have from the above equations,

$$\frac{jH_0\beta}{\rho}e^{j\beta z}\mathbf{a}_\phi = j\omega\varepsilon_0\frac{50}{\rho}e^{j\beta z}\mathbf{a}_\phi$$

or

$$H_0\beta = 50\omega\varepsilon_0$$

$$-j\beta\frac{50}{\rho}e^{j\beta z}\mathbf{a}_\rho = -j\omega\mu_0\frac{H_0}{\rho}e^{j\beta z}\mathbf{a}_\rho$$

or

$$\frac{H_0}{\beta} = \frac{50}{\omega\mu_0}$$

$$H_0^2 = (50)^2\frac{\varepsilon_0}{\mu_0}$$

or

$$H_0 = \pm 50\sqrt{\frac{\varepsilon_0}{\mu_0}} = \pm \frac{50}{120\pi} = \pm 0.1326$$

$$\beta^2 = \omega^2\mu_0\varepsilon_0$$

or

$$\beta = \pm\omega\sqrt{\mu_0\varepsilon_0} = \pm\frac{\omega}{c} = \pm\frac{10^6}{3 \times 10^8}$$

$$= \pm 3.33 \times 10^{-3}$$

$$H_0 = 0.1326, \beta = 3.33 \times 10^{-3}$$

or

$$H_0 = -0.1326, \beta = -3.33 \times 10^{-3};$$

only these will satisfy Maxwell's four equations.

Example 7:

In a medium characterized by

$$\sigma = 0, \mu = \mu_0, \varepsilon = 4\varepsilon_0,$$

and

$$\mathbf{E} = 20 \sin(10^8 t - \beta z) \mathbf{a}_y \text{ V/m}$$

calculate β and $\mathbf{H}$.

Solution:

Method 1 (time domain):

Let us solve this problem the harder way—in time domain. It is evident that Gauss's law for electric fields is satisfied; that is,

$$\nabla \cdot \mathbf{E} = \frac{\partial E_y}{\partial y} = 0$$

From Faraday's law,

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \rightarrow \mathbf{H} = -\frac{1}{\mu} \int (\nabla \times \mathbf{E}) dt$$

But

$$\begin{aligned} \nabla \times \mathbf{E} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_y & 0 \end{vmatrix} = -\frac{\partial E_y}{\partial z} \mathbf{a}_x + \frac{\partial E_y}{\partial x} \mathbf{a}_z \\ &= 20\beta \cos(10^8 t - \beta z) \mathbf{a}_x + 0 \end{aligned}$$

Hence,

$$\begin{aligned} \mathbf{H} &= -\frac{20\beta}{\mu} \int \cos(10^8 t - \beta z) dt \mathbf{a}_x \\ &= -\frac{20\beta}{\mu 10^8} \sin(10^8 t - \beta z) \mathbf{a}_x \end{aligned}$$

It is readily verified that

$$\nabla \cdot \mathbf{H} = \frac{\partial H_x}{\partial x} = 0$$

showing that Gauss's law for magnetic fields is satisfied. Lastly, from Ampère's law

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \rightarrow \mathbf{E} = \frac{1}{\varepsilon} \int (\nabla \times \mathbf{H}) dt$$

because $\sigma = 0$.

But

$$\begin{aligned} \nabla \times \mathbf{H} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & 0 & 0 \end{vmatrix} = \frac{\partial H_x}{\partial z} \mathbf{a}_y - \frac{\partial H_x}{\partial y} \mathbf{a}_z \\ &= \frac{20\beta^2}{\mu 10^8} \cos(10^8 t - \beta z) \mathbf{a}_y + 0 \end{aligned}$$

where $\mathbf{H}$ in the above eq. has been substituted. Thus,

$$\begin{aligned}\mathbf{E} &= \frac{20\beta^2}{\mu\varepsilon 10^8} \int \cos(10^8 t - \beta z) dt \mathbf{a}_y \\ &= \frac{20\beta^2}{\mu\varepsilon 10^{16}} \sin(10^8 t - \beta z) \mathbf{a}_y\end{aligned}$$

Comparing this with the given $\mathbf{E}$, we have

$$\frac{20\beta^2}{\mu\varepsilon 10^{16}} = 20$$

or

$$\begin{aligned}\beta &= \pm 10^8 \sqrt{\mu\varepsilon} = \pm 10^8 \sqrt{\mu_0 \cdot 4\varepsilon_0} = \pm \frac{10^8(2)}{c} = \pm \frac{10^8(2)}{3 \times 10^8} \\ &= \pm \frac{2}{3}\end{aligned}$$

The β would be negative only in metamaterials, for an isotropic medium,

$$\beta = \frac{2}{3}$$

$$\mathbf{H} = + \frac{20(2/3)}{4\pi \cdot 10^{-7}(10^8)} \sin\left(10^8 t - \frac{2z}{3}\right) \mathbf{a}_x$$

or

$$\mathbf{H} = + \frac{1}{3\pi} \sin\left(10^8 t - \frac{2z}{3}\right) \mathbf{a}_x \text{ A/m}$$

Method 2 (using phasors):

$$\mathbf{E} = \text{Im}(\mathbf{E}_s e^{j\omega t}) \rightarrow \mathbf{E}_s = 20e^{-j\beta z} \mathbf{a}_y$$

where $\omega = 10^8$.

Again

$$\nabla \cdot \mathbf{E}_s = \frac{\partial E_{ys}}{\partial y} = 0$$

$$\nabla \times \mathbf{E}_s = -j\omega\mu\mathbf{H}_s \rightarrow \mathbf{H}_s = \frac{\nabla \times \mathbf{E}_s}{-j\omega\mu}$$

$$\mathbf{H}_s = \frac{1}{-j\omega\mu} \left[-\frac{\partial E_{ys}}{\partial z} \mathbf{a}_x \right] = \frac{-20\beta}{\omega\mu} e^{-j\beta z} \mathbf{a}_x$$

Notice that $\nabla \cdot \mathbf{H}_s = 0$ is satisfied.

$$\nabla \times \mathbf{H}_s = j\omega\varepsilon\mathbf{E}_s \rightarrow \mathbf{E}_s = \frac{\nabla \times \mathbf{H}_s}{j\omega\varepsilon}$$

$$\mathbf{E}_s = \frac{1}{j\omega\epsilon} \frac{\partial H_{xs}}{\partial z} \mathbf{a}_y = \frac{20\beta^2 e^{-j\beta z}}{\omega^2 \mu \epsilon} \mathbf{a}_y$$

Comparing this with the given $\mathbf{E}$ in previous equation,

$$20 = \frac{20\beta^2}{\omega^2 \mu \epsilon}$$

or

$$\beta = +\omega\sqrt{\mu\epsilon} = \frac{2}{3}$$

as obtained before.

$$\mathbf{H}_s = +\frac{20(2/3)}{10^8(4\pi \times 10^{-7})} \mathbf{a}_x = +\frac{1}{3\pi} e^{-j\beta z} \mathbf{a}_x$$

$$\mathbf{H} = \text{Im}(\mathbf{H}_s e^{j\omega t})$$

$$= \pm \frac{1}{3\pi} \sin(10^8 t - \beta z) \mathbf{a}_x \text{ A/m}$$

as obtained before. It should be noticed that working with phasors is considerably simpler than working directly in time domain. Also, notice that we have used

$$\mathbf{A} = \text{Im}(\mathbf{A}_s e^{j\omega t})$$

because the given $\mathbf{E}$ is in sine form and not cosine. If we had used

$$\mathbf{A} = \text{Re}(\mathbf{A}_s e^{j\omega t})$$

sine would have been expressed in terms of cosine,

$$\mathbf{E} = 20 \cos(10^8 t - \beta z - 90^\circ) \mathbf{a}_y = \text{Re}(\mathbf{E}_s e^{j\omega t})$$

or

$$\mathbf{E}_s = 20 e^{-j\beta z - j90^\circ} \mathbf{a}_y = -j20 e^{-j\beta z} \mathbf{a}_y$$

and we follow the same procedure.

Example 8:

An electric field in free space is given by

$$\mathbf{E} = 50 \cos(10^8 t + \beta x) \mathbf{a}_y \text{ V/m}$$

- Find the direction of wave propagation.
- Calculate β and the time it takes to travel a distance of $\lambda/2$.
- Sketch the wave at $t = 0$, $T/4$, and $T/2$.

Solution:

(a) From the positive sign in $(\omega t + \beta x)$, we infer that the wave is propagating along $-\mathbf{a}_x$. This will be confirmed in part (c) of this example.

(b) In free space, $u = c$:

$$\beta = \frac{\omega}{c} = \frac{10^8}{3 \times 10^8} = \frac{1}{3}$$

or

$$\beta = 0.3333 \text{ rad/m}$$

If T is the period of the wave, it takes T seconds to travel a distance λ at speed c . Hence to travel a distance of $\lambda/2$ will take

$$t_1 = \frac{T}{2} = \frac{1}{2} \frac{2\pi}{\omega} = \frac{\pi}{10^8} = 31.42 \text{ ns}$$

Alternatively, because the wave is traveling at the speed of light c ,

$$\frac{\lambda}{2} = ct_1 \quad \text{or} \quad t_1 = \frac{\lambda}{2c}$$

But

$$\lambda = \frac{2\pi}{\beta} = 6\pi$$

Hence,

$$t_1 = \frac{6\pi}{2(3 \times 10^8)} = 31.42 \text{ ns}$$

as obtained before.

(c) At

$$t = 0, \quad E_y = 50 \cos \beta x$$

At

$$\begin{aligned} t = T/4, \quad E_y &= 50 \cos \left(\omega \cdot \frac{2\pi}{4\omega} + \beta x \right) = 50 \cos \left(\beta x + \frac{\pi}{2} \right) \\ &= -50 \sin \beta x \end{aligned}$$

At

$$\begin{aligned}
 t = T/2, \quad E_y &= 50 \cos \left(\omega \cdot \frac{2\pi}{2\omega} + \beta x \right) = 50 \cos(\beta x + \pi) \\
 &= -50 \cos \beta x
 \end{aligned}$$

E_y at $t = 0, T/4, T/2$ is plotted against x as shown in Figure 4. Notice that a point P (arbitrarily selected) on the wave moves along $-\mathbf{a}_x$ as t increases with time. This shows that the wave travels along $-\mathbf{a}_x$.

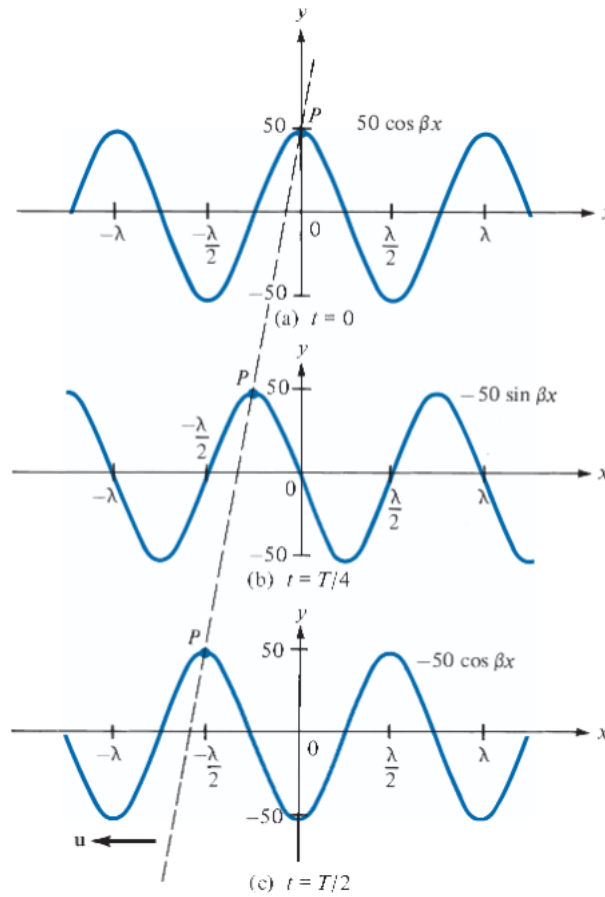


Figure 4: wave travels along $-\mathbf{a}_x$

Example 9:

A lossy dielectric has an intrinsic impedance of $200\angle 30^\circ \Omega$ at a particular radian frequency ω . If, at that frequency, the plane wave propagating through the dielectric has the magnetic field component

$$\mathbf{H} = 10e^{-\alpha x} \cos\left(\omega t - \frac{1}{2}x\right) \mathbf{a}_y \text{ A/m}$$

find $\mathbf{E}$ and α .

Solution:

The given wave travels along $\mathbf{a}_x$ so that $\mathbf{a}_k = \mathbf{a}_x$; $\mathbf{a}_H = \mathbf{a}_y$, so

$$-\mathbf{a}_E = \mathbf{a}_k \times \mathbf{a}_H = \mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z$$

or

$$\mathbf{a}_E = -\mathbf{a}_z$$

Also $H_o = 10$, so

$$\frac{E_o}{H_o} = \eta = 200 \angle 30^\circ = 200e^{j\pi/6} \rightarrow E_o = 2000e^{j\pi/6}$$

Except for the amplitude and phase difference, $\mathbf{E}$ and $\mathbf{H}$ always have the same form. Hence

$$\mathbf{E} = \text{Re}\left(2000e^{j\pi/6}e^{-\gamma x}e^{j\omega t}\mathbf{a}_E\right)$$

or

$$\mathbf{E} = -2e^{-\alpha x} \cos\left(\omega t - \frac{x}{2} + \frac{\pi}{6}\right) \mathbf{a}_z \text{ kV/m}$$

Knowing that $\beta = 1/2$, we need to determine α . Since

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1 \right]}$$

and

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1 \right]}$$

$$\frac{\alpha}{\beta} = \left[\frac{\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1}{\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1} \right]^{1/2}$$

But

$$\frac{\sigma}{\omega\varepsilon} = \tan 2\theta_\eta = \tan 60^\circ = \sqrt{3}$$

Hence,

$$\frac{\alpha}{\beta} = \left[\frac{2-1}{2+1} \right]^{1/2} = \frac{1}{\sqrt{3}}$$

or

$$\alpha = \frac{\beta}{\sqrt{3}} = \frac{1}{2\sqrt{3}} = 0.2887 \text{ Np/m}$$

Example 10:

In a lossless dielectric for which $\eta = 60\pi$, $\mu_r = 1$, and

$$\mathbf{H} = -0.1 \cos(\omega t - z) \mathbf{a}_x + 0.5 \sin(\omega t - z) \mathbf{a}_y \text{ A/m},$$

calculate ε_r , ω , and $\mathbf{E}$.

Solution:

In this case, $\sigma = 0$, $\alpha = 0$, and $\beta = 1$, so

$$\eta = \sqrt{\mu/\varepsilon} = \sqrt{\frac{\mu_o}{\varepsilon_o} \sqrt{\frac{\mu_r}{\varepsilon_r}}} = \frac{120\pi}{\sqrt{\varepsilon_r}}$$

or

$$\sqrt{\varepsilon_r} = \frac{120\pi}{\eta} = \frac{120\pi}{60\pi} = 2 \quad \rightarrow \quad \varepsilon_r = 4$$

$$\beta = \omega \sqrt{\mu\varepsilon} = \omega \sqrt{\mu_o \varepsilon_o} \sqrt{\mu_r \varepsilon_r} = \frac{\omega}{c} \sqrt{4} = \frac{2\omega}{c}$$

or

$$\omega = \frac{\beta c}{2} = \frac{1(3 \times 10^8)}{2} = 1.5 \times 10^8 \text{ rad/s}$$

From the given $\mathbf{H}$ field, $\mathbf{E}$ can be calculated in two ways: by using the techniques (based on Maxwell's equations) developed in this chapter or directly, by using Maxwell's equations.

Method 1:

To use the techniques developed in the present chapter, we let

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2$$

where

$$\mathbf{H}_1 = -0.1 \cos(\omega t - z) \mathbf{a}_x$$

and

$$\mathbf{H}_2 = 0.5 \sin(\omega t - z) \mathbf{a}_y$$

and the corresponding electric field

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$$

where

$$\mathbf{E}_1 = E_{1o} \cos(\omega t - z) \mathbf{a}_y$$

and

$$\mathbf{E}_2 = E_{2o} \sin(\omega t - z) \mathbf{a}_x$$

Notice that although $\mathbf{H}$ has components along $\mathbf{a}_x$ and $\mathbf{a}_y$, it has no component along the direction of propagation; it is therefore a TEM wave.

For $\mathbf{E}_1$,

$$\mathbf{a}_{E_1} = -(\mathbf{a}_k \times \mathbf{a}_{H_1}) = -(\mathbf{a}_z \times \mathbf{a}_x) = \mathbf{a}_y$$

$$E_{1o} = \eta H_{1o} = 60\pi(0.1) = 6\pi$$

Hence

$$\mathbf{E}_1 = 6\pi \cos(\omega t - z) \mathbf{a}_y$$

For $\mathbf{E}_2$,

$$\mathbf{a}_{E_2} = -(\mathbf{a}_k \times \mathbf{a}_{H_2}) = -(\mathbf{a}_z \times \mathbf{a}_y) = \mathbf{a}_x$$

$$E_{2o} = \eta H_{2o} = 60\pi(0.5) = 30\pi$$

Hence

$$\mathbf{E}_2 = 30\pi \sin(\omega t - z) \mathbf{a}_x$$

Adding $\mathbf{E}_1$ and $\mathbf{E}_2$ gives $\mathbf{E}$; that is,

$$\mathbf{E} = 94.25 \sin(1.5 \times 10^8 t - z) \mathbf{a}_x + 18.85 \cos(1.5 \times 10^8 t - z) \mathbf{a}_y \text{ V/m}$$

Method 2:

We may apply Maxwell's equations directly

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad \rightarrow \quad \mathbf{E} = \frac{1}{\varepsilon} \int \nabla \times \mathbf{H} dt$$

because $\sigma = 0$. But

$$\begin{aligned}\nabla \times \mathbf{H} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x(z) & H_y(z) & 0 \end{vmatrix} = -\frac{\partial H_y}{\partial z} \mathbf{a}_x + \frac{\partial H_x}{\partial z} \mathbf{a}_y \\ &= H_{2o} \cos(\omega t - z) \mathbf{a}_x + H_{1o} \sin(\omega t - z) \mathbf{a}_y\end{aligned}$$

where $H_{1o} = -0.1$ and $H_{2o} = 0.5$. Hence

$$\begin{aligned}\mathbf{E} &= \frac{1}{\varepsilon} \int \nabla \times \mathbf{H} dt = \frac{H_{2o}}{\varepsilon \omega} \sin(\omega t - z) \mathbf{a}_x - \frac{H_{1o}}{\varepsilon \omega} \cos(\omega t - z) \mathbf{a}_y \\ &= 94.25 \sin(\omega t - z) \mathbf{a}_x + 18.85 \cos(\omega t - z) \mathbf{a}_y, \text{ V/m}\end{aligned}$$

as expected.

Example 11:

A uniform plane wave propagating in a medium has

$$\mathbf{E} = 2e^{-\alpha z} \sin(10^8 t - \beta z) \mathbf{a}_y \text{ V/m}$$

If the medium is characterized by $\varepsilon_r = 1$, $\mu_r = 20$, and $\sigma = 3 \text{ S/m}$, find α , β , and $\mathbf{H}$.

Solution:

We need to determine the loss tangent to be able to tell whether the medium is a lossy dielectric or a good conductor.

$$\frac{\sigma}{\omega \varepsilon} = \frac{3}{10^8 \times 1 \times \frac{10^{-9}}{36\pi}} = 3393 \gg 1$$

showing that the medium may be regarded as a good conductor at the frequency of operation. Hence,

$$\alpha = \beta = \sqrt{\frac{\mu \omega \sigma}{2}} = \left[\frac{4\pi \times 10^{-7} \times 20(10^8)(3)}{2} \right]^{1/2} = 61.4$$

$$\alpha = 61.4 \text{ Np/m}, \quad \beta = 61.4 \text{ rad/m}$$

Also

$$|\eta| = \sqrt{\frac{\mu \omega}{\sigma}} = \left[\frac{4\pi \times 10^{-7} \times 20(10^8)}{3} \right]^{1/2} = \sqrt{\frac{800\pi}{3}}$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega \varepsilon} = 3393 \rightarrow \theta_\eta = 45^\circ = \frac{\pi}{4}$$

Hence

$$\mathbf{H} = H_o e^{-\alpha z} \sin\left(\omega t - \beta z - \frac{\pi}{4}\right) \mathbf{a}_H$$

where

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_z \times \mathbf{a}_y = -\mathbf{a}_x$$

and

$$H_o = \frac{E_o}{|\eta|} = 2\sqrt{\frac{3}{800\pi}} = 69.1 \times 10^{-3}$$

Thus

$$\mathbf{H} = -69.1 e^{-61.4z} \sin\left(10^8 t - 61.4z - \frac{\pi}{4}\right) \mathbf{a}_x \text{ mA/m}$$

Example 12:

A plane wave

$$\mathbf{E} = E_o \cos(\omega t - \beta z) \mathbf{a}_x$$

is incident on a good conductor at $z \geq 0$. Find the current density in the conductor.

Solution:

Since the current density $\mathbf{J} = \sigma \mathbf{E}$, we expect $\mathbf{J}$ to satisfy the wave equation.

$$\nabla^2 \mathbf{J}_s - \gamma^2 \mathbf{J}_s = 0$$

Also the incident $\mathbf{E}$ has only an x -component and varies with z . Hence $\mathbf{J} = J_x(z, t) \mathbf{a}_x$ and

$$\frac{d^2}{dz^2} J_x - \gamma^2 J_x = 0$$

which is an ordinary differential equation with solution

$$J_x = A e^{-\gamma z} + B e^{+\gamma z}$$

The constant B must be zero because J_x is finite as $z \rightarrow \infty$. But in a good conductor,

$$\sigma \gg \omega \varepsilon, \quad \text{so that} \quad \alpha = \beta = \frac{1}{\delta}$$

Hence

$$\gamma = \alpha + j\beta = \alpha(1 + j) = \frac{(1 + j)}{\delta}$$

and

$$J_x = Ae^{-z(1+j)/\delta}$$

or

$$J_x = J_{sx}(0) e^{-z(1+j)/\delta}$$

where $J_{sx}(0)$ is the current density on the conductor surface.

Example 13:

Determine the polarization of a plane wave with:

$$(a) \quad \mathbf{E}(z, t) = 4e^{-0.25z} \cos(\omega t - 0.8z) \mathbf{a}_x + 3e^{-0.25z} \sin(\omega t - 0.8z) \mathbf{a}_y \text{ V/m}$$

$$(b) \quad \mathbf{H}_s(z) = H_o e^{-j\beta z} \mathbf{a}_x - 2H_o e^{-j\beta z} \mathbf{a}_y$$

Solution:

(a)

From the given $\mathbf{E}$,

$$E_x = 4e^{-0.25z} \cos(\omega t - 0.8z)$$

$$E_y = 3e^{-0.25z} \sin(\omega t - 0.8z)$$

In the $z = 0$ plane, we have

$$\frac{1}{4} E_x(0, t) = \cos(\omega t)$$

$$\frac{1}{3} E_y(0, t) = \sin(\omega t)$$

Squaring and adding gives

$$\frac{1}{16} E_x^2(0, t) + \frac{1}{9} E_y^2(0, t) = 1$$

which describes an ellipse. Hence, the wave is elliptically polarized.

(b)

The two components of $\mathbf{H}$ are in phase; hence, the polarization is linear. For proper characterization, it is expedient to find the electric field component. This can be done in many ways. Using Maxwell's equation,

$$\nabla \times \mathbf{H}_s = j\omega\epsilon \mathbf{E}_s \quad \longrightarrow \quad \mathbf{E}_s = \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H}_s$$

$$\nabla \times \mathbf{H}_s = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_o e^{-j\beta z} & -2H_o e^{-j\beta z} & 0 \end{vmatrix} = -j2\beta H_o e^{-j\beta z} \mathbf{a}_x - j\beta H_o e^{-j\beta z} \mathbf{a}_y$$

Dividing both sides by $j\omega\varepsilon$ and setting $\eta = \beta/\omega\varepsilon$ yields

$$\mathbf{E}_s = 2\eta H_o e^{-j\beta z} \mathbf{a}_x - \eta H_o e^{-j\beta z} \mathbf{a}_y$$

In the time domain,

$$\mathbf{E}(z, t) = \text{Re} [\mathbf{E}_s e^{j\omega t}] = 2\eta H_o \cos(\omega t - \beta z) \mathbf{a}_x - \eta H_o \cos(\omega t - \beta z) \mathbf{a}_y$$

If we set $z = 0$,

$$\mathbf{E}(0, t) = (2\eta H_o \mathbf{a}_x - \eta H_o \mathbf{a}_y) \cos(\omega t)$$

At $t = 0$, $\mathbf{E}$ has components $2\eta H_o$ in the x -direction and ηH_o in the y -direction. The ratio E_y/E_x remains the same as t changes. Hence, $\mathbf{E}$ is linearly polarized.

Example 14:

In a nonmagnetic medium

$$\mathbf{E} = 4 \sin(2\pi \times 10^7 t - 0.8x) \mathbf{a}_z \text{ V/m}$$

Find

- (a) ε_r , η
- (b) The time-average power carried by the wave
- (c) The total power crossing 100 cm^2 of plane $2x + y = 5$

Solution:

- (a) Since $\alpha = 0$ and $\beta \neq \omega/c$, the medium is not free space but a lossless medium:

$$\beta = 0.8, \quad \omega = 2\pi \times 10^7, \quad \mu = \mu_o \text{ (nonmagnetic)}, \quad \varepsilon = \varepsilon_o \varepsilon_r$$

Hence

$$\beta = \omega \sqrt{\mu \varepsilon} = \omega \sqrt{\mu_o \varepsilon_o \varepsilon_r} = \frac{\omega}{c} \sqrt{\varepsilon_r}$$

or

$$\sqrt{\varepsilon_r} = \frac{\beta c}{\omega} = \frac{0.8(3 \times 10^8)}{2\pi \times 10^7} = \frac{12}{\pi}$$

$$\varepsilon_r = 14.59$$

$$\begin{aligned}\eta &= \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_o}{\varepsilon_o \varepsilon_r}} = \frac{120\pi}{\sqrt{\varepsilon_r}} = 120\pi \cdot \frac{\pi}{12} = 10\pi^2 \\ &= 98.7 \, \Omega\end{aligned}$$

(b)

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} = \frac{E_o^2}{\eta} \sin^2(\omega t - \beta x) \mathbf{a}_x$$

$$\begin{aligned}\mathcal{P}_{ave} &= \frac{1}{T} \int_0^T \mathcal{P} dt = \frac{E_o^2}{2\eta} \mathbf{a}_x = \frac{16}{2 \times 10\pi^2} \mathbf{a}_x \\ &= 81 \mathbf{a}_x \text{ mW/m}^2\end{aligned}$$

(c) On plane $2x + y = 5$,

$$\mathbf{a}_n = \frac{2\mathbf{a}_x + \mathbf{a}_y}{\sqrt{5}}$$

Hence the total power is

$$\begin{aligned}P_{ave} &= \int \mathcal{P}_{ave} \cdot d\mathbf{S} = \mathcal{P}_{ave} \cdot S \mathbf{a}_n \\ &= (81 \times 10^{-3} \mathbf{a}_x) \cdot (100 \times 10^{-4}) \left[\frac{2\mathbf{a}_x + \mathbf{a}_y}{\sqrt{5}} \right] \\ &= \frac{162 \times 10^{-5}}{\sqrt{5}} = 724.5 \, \mu\text{W}\end{aligned}$$

Example 15:

In free space ($z \leq 0$), a plane wave with

$$\mathbf{H}_i = 10 \cos(10^8 t - \beta z) \mathbf{a}_x \text{ mA/m}$$

is incident normally on a lossless medium ($\varepsilon = 2\varepsilon_o$, $\mu = 8\mu_o$) in region $z \geq 0$. Determine the reflected wave $\mathbf{H}_r$, $\mathbf{E}_r$, and the transmitted wave $\mathbf{H}_t$, $\mathbf{E}_t$.

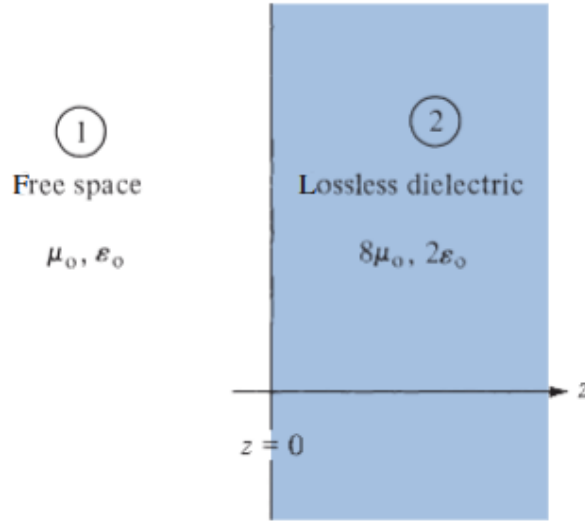


Figure 5: Illustration of the problem

Solution:

This problem can be solved in two different ways.

Method 1:

Consider the problem as illustrated in Figure 5. For free space,

$$\beta_1 = \frac{\omega}{c} = \frac{10^8}{3 \times 10^8} = \frac{1}{3}$$

$$\eta_1 = \eta_0 = 120\pi$$

For the lossless dielectric medium,

$$\beta_2 = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\mu_0\epsilon_0}\sqrt{\mu_r\epsilon_r} = \frac{\omega}{c}(4) = 4\beta_1 = \frac{4}{3}$$

$$\eta_2 = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0} \frac{\mu_r}{\epsilon_r}} = 2\eta_0$$

Given that

$$\mathbf{H}_i = 10 \cos(10^8 t - \beta_1 z) \mathbf{a}_x \text{ mA/m},$$

we expect that

$$\mathbf{E}_i = E_{io} \cos(10^8 t - \beta_1 z) \mathbf{a}_{E_i}$$

where

$$\mathbf{a}_{E_i} = \mathbf{a}_{H_i} \times \mathbf{a}_k = \mathbf{a}_x \times \mathbf{a}_z = -\mathbf{a}_y$$

and

$$E_{io} = \eta_1 H_{io} = 10\eta_o$$

Hence,

$$\mathbf{E}_i = -10\eta_o \cos(10^8 t - \beta_1 z) \mathbf{a}_y \text{ mV/m}$$

Now

$$\frac{E_{ro}}{E_{io}} = \Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{2\eta_o - \eta_o}{2\eta_o + \eta_o} = \frac{1}{3}$$

$$E_{ro} = \frac{1}{3} E_{io}$$

Thus

$$\mathbf{E}_r = -\frac{10}{3} \eta_o \cos\left(10^8 t + \frac{1}{3} z\right) \mathbf{a}_y \text{ mV/m}$$

from which we easily obtain $\mathbf{H}_r$ as

$$\mathbf{H}_r = -\frac{10}{3} \cos\left(10^8 t + \frac{1}{3} z\right) \mathbf{a}_x \text{ mA/m}$$

Similarly,

$$\frac{E_{to}}{E_{io}} = \tau = 1 + \Gamma = \frac{4}{3}$$

or

$$E_{to} = \frac{4}{3} E_{io}$$

Thus

$$\mathbf{E}_t = E_{to} \cos(10^8 t - \beta_2 z) \mathbf{a}_{E_t}$$

where

$$\mathbf{a}_{E_t} = -\mathbf{a}_y$$

Hence,

$$\mathbf{E}_t = -\frac{40}{3} \eta_o \cos\left(10^8 t - \frac{4}{3} z\right) \mathbf{a}_y \text{ mV/m}$$

from which we obtain

$$\mathbf{H}_t = \frac{20}{3} \cos\left(10^8 t - \frac{4}{3} z\right) \mathbf{a}_x \text{ mA/m}$$

Method 2:

Alternatively, we can obtain $\mathbf{H}_r$ and $\mathbf{H}_t$ directly from $\mathbf{H}_i$ by using

$$\frac{H_{ro}}{H_{io}} = -\Gamma \quad \text{and} \quad \frac{H_{to}}{H_{io}} = \tau \frac{\eta_1}{\eta_2}$$

Thus

$$H_{ro} = -\frac{1}{3}H_{io} = -\frac{10}{3}$$

$$H_{to} = \frac{4}{3} \frac{\eta_o}{2\eta_o} H_{io} = \frac{2}{3} H_{io} = \frac{20}{3}$$

and

$$\mathbf{H}_r = -\frac{10}{3} \cos(10^8 t + \beta_1 z) \mathbf{a}_x \text{ mA/m}$$

$$\mathbf{H}_t = \frac{20}{3} \cos(10^8 t - \beta_2 z) \mathbf{a}_x \text{ mA/m}$$

as obtained by Method 1.

Notice that the boundary conditions at $z = 0$, namely,

$$\mathbf{E}_i(0) + \mathbf{E}_r(0) = \mathbf{E}_t(0) = -\frac{40}{3} \eta_o \cos(10^8 t) \mathbf{a}_y$$

and

$$\mathbf{H}_i(0) + \mathbf{H}_r(0) = \mathbf{H}_t(0) = \frac{20}{3} \cos(10^8 t) \mathbf{a}_x$$

are satisfied. The boundary conditions can always be used to cross-check $\mathbf{E}$ and $\mathbf{H}$.

Example 16:

Given a uniform plane wave in air as

$$\mathbf{E}_i = 40 \cos(\omega t - \beta z) \mathbf{a}_x + 30 \sin(\omega t - \beta z) \mathbf{a}_y \text{ V/m}$$

- (a) Find $\mathbf{H}_i$.
- (b) If the wave encounters a perfectly conducting plate normal to the z -axis at $z = 0$, find the reflected wave $\mathbf{E}_r$ and $\mathbf{H}_r$.
- (c) What are the total $\mathbf{E}$ and $\mathbf{H}$ fields for $z \leq 0$?
- (d) Calculate the time-average Poynting vectors for $z \leq 0$ and $z \geq 0$.

Solution:

(a) We may treat the wave as consisting of two waves $\mathbf{E}_{i1}$ and $\mathbf{E}_{i2}$, where

$$\mathbf{E}_{i1} = 40 \cos(\omega t - \beta z) \mathbf{a}_x, \quad \mathbf{E}_{i2} = 30 \sin(\omega t - \beta z) \mathbf{a}_y$$

At atmospheric pressure, air has $\varepsilon_r = 1.0006 = 1$. Thus air may be regarded as free space.

Let

$$\mathbf{H}_i = \mathbf{H}_{i1} + \mathbf{H}_{i2}$$

$$\mathbf{H}_{i1} = H_{1o} \cos(\omega t - \beta z) \mathbf{a}_{H1}$$

where

$$H_{1o} = \frac{E_{1o}}{\eta_o} = \frac{40}{120\pi} = \frac{1}{3\pi}$$

$$\mathbf{a}_{H1} = \mathbf{a}_k \times \mathbf{a}_{E1} = \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$

Hence

$$\mathbf{H}_{i1} = \frac{1}{3\pi} \cos(\omega t - \beta z) \mathbf{a}_y$$

Similarly,

$$\mathbf{H}_{i2} = H_{2o} \sin(\omega t - \beta z) \mathbf{a}_{H2}$$

where

$$H_{2o} = \frac{E_{2o}}{\eta_o} = \frac{30}{120\pi} = \frac{1}{4\pi}$$

$$\mathbf{a}_{H2} = \mathbf{a}_k \times \mathbf{a}_{E2} = \mathbf{a}_z \times \mathbf{a}_y = -\mathbf{a}_x$$

Hence

$$\mathbf{H}_{i2} = -\frac{1}{4\pi} \sin(\omega t - \beta z) \mathbf{a}_x$$

and

$$\mathbf{H}_i = \mathbf{H}_{i1} + \mathbf{H}_{i2} = -\frac{1}{4\pi} \sin(\omega t - \beta z) \mathbf{a}_x + \frac{1}{3\pi} \cos(\omega t - \beta z) \mathbf{a}_y \text{ mA/m}$$

(b) Since medium 2 is perfectly conducting,

$$\frac{\sigma_2}{\omega\epsilon_2} \gg 1 \quad \rightarrow \quad \eta_2 \ll \eta_1$$

that is,

$$\Gamma = -1, \quad \tau = 0$$

showing that the incident $\mathbf{E}$ and $\mathbf{H}$ fields are totally reflected:

$$E_{ro} = \Gamma E_{io} = -E_{io}$$

Hence,

$$\mathbf{E}_r = -40 \cos(\omega t + \beta z) \mathbf{a}_x - 30 \sin(\omega t + \beta z) \mathbf{a}_y \text{ V/m}$$

We can find $\mathbf{H}_r$ from $\mathbf{E}_r$ just as we did in part (a) of this example. Whichever approach is taken, we obtain

$$\mathbf{H}_r = \frac{1}{3\pi} \cos(\omega t + \beta z) \mathbf{a}_y - \frac{1}{4\pi} \sin(\omega t + \beta z) \mathbf{a}_x \text{ A/m}$$

(c) The total fields in air

$$\mathbf{E}_1 = \mathbf{E}_i + \mathbf{E}_r \quad \text{and} \quad \mathbf{H}_1 = \mathbf{H}_i + \mathbf{H}_r$$

can be shown to be standing waves. The total fields in the conductor are

$$\mathbf{E}_2 = \mathbf{E}_t = 0, \quad \mathbf{H}_2 = \mathbf{H}_t = 0$$

(d) For $z \leq 0$,

$$\begin{aligned} \mathcal{P}_{1ave} &= \frac{|\mathbf{E}_{1s}|^2}{2\eta_1} \mathbf{a}_k = \frac{1}{2\eta_o} [E_{io}^2 \mathbf{a}_z - E_{ro}^2 \mathbf{a}_z] \\ &= \frac{1}{240\pi} [(40^2 + 30^2) \mathbf{a}_z - (40^2 + 30^2) \mathbf{a}_z] \\ &= 0 \end{aligned}$$

For $z \geq 0$,

$$\mathcal{P}_{2ave} = \frac{|\mathbf{E}_{2s}|^2}{2\eta_2} \mathbf{a}_k = \frac{E_{to}^2}{2\eta_2} \mathbf{a}_z = 0$$

Example 17:

An EM wave travels in free space with the electric field component

$$\mathbf{E}_s = 100 e^{j(0.866y+0.5z)} \mathbf{a}_x \text{ V/m}$$

Determine

- (a) ω and λ
- (b) The magnetic field component
- (c) The time average power in the wave

Solution:

- (a) Comparing the given $\mathbf{E}$ with

$$\mathbf{E}_s = E_o e^{j\mathbf{k} \cdot \mathbf{r}} = E_o e^{j(k_x x + k_y y + k_z z)} \mathbf{a}_x$$

it is clear that

$$k_x = 0, \quad k_y = 0.866, \quad k_z = 0.5$$

Thus

$$k = \sqrt{k_x^2 + k_y^2 + k_z^2} = \sqrt{(0.866)^2 + (0.5)^2} = 1$$

But in free space,

$$k = \beta = \omega \sqrt{\mu_o \epsilon_o} = \frac{\omega}{c} = \frac{2\pi}{\lambda}$$

Hence,

$$\omega = kc = 3 \times 10^8 \text{ rad/s}$$

$$\lambda = \frac{2\pi}{k} = 2\pi = 6.283 \text{ m}$$

(b) The corresponding magnetic field is given by

$$\begin{aligned}\mathbf{H}_s &= \frac{1}{\mu\omega} \mathbf{k} \times \mathbf{E}_s \\ &= \frac{(0.866\mathbf{a}_y + 0.5\mathbf{a}_z)}{4\pi \times 10^{-7} \times 3 \times 10^8} \times 100 \mathbf{a}_x e^{j\mathbf{k} \cdot \mathbf{r}}\end{aligned}$$

or

$$\mathbf{H}_s = (132.63\mathbf{a}_y - 229.7\mathbf{a}_z) e^{j(0.866y + 0.5z)} \text{ mA/m}$$

(c) The time-average power is

$$\begin{aligned}\mathcal{P}_{ave} &= \frac{1}{2} \text{Re}(\mathbf{E}_s \times \mathbf{H}_s^*) = \frac{E_o^2}{2\eta} \mathbf{a}_k \\ &= \frac{(100)^2}{2(120\pi)} (0.866\mathbf{a}_y + 0.5\mathbf{a}_z) \\ &= 11.49\mathbf{a}_y + 6.631\mathbf{a}_z \text{ W/m}^2\end{aligned}$$